

A Rederivation of the AYO ExoEarth Yield Formalism, and HWO Yields for a Mass-Selected Planet Sample in a Narrowed Habitable Zone

Prepared in the `yields` vaibify project

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Abstract

This report rederives the Altruistic Yield Optimization (AYO) exoEarth-candidate yield formalism of Stark et al. (2014, 2019, 2024) from the published equations and parameter tables, and uses it to answer a counterfactual: what would the Habitable Worlds Observatory (HWO) baseline yield if the planet sample were defined by mass, $0.5\text{--}2\,M_{\oplus}$ with $R = M^{0.27}$, and if the habitable zone were 0.96–1.20 AU for the Sun rather than the conservative 0.95–1.67 AU, scaled by $\sqrt{L_{\star}}$ for every star. Integrating the SAG13 occurrence law over the redefined box gives $\eta = 0.0520^{+0.0345}_{-0.0207}$ against $0.256^{+0.168}_{-0.101}$ for the canonical box, a ratio of $0.203^{+0.017}_{-0.017}$ that is independent of the overall normalisation. The full survey optimization gives a *yield* ratio of $0.210^{+0.025}_{-0.021}$, materially larger than the occurrence ratio, because a smaller expected detection count frees characterization time and because truncating the outer habitable zone preferentially removes the faintest, most expensive planets. The baseline 6 m design yields a median of 4.7 exoEarth candidates under the redefined selection, against 22.7 canonically, and the probability of reaching the 25-planet goal falls from 43.3% to 0.19%. An out-of-sample test against the published aperture scaling (Sec. 4.5) drove two corrections to the characterization model, after which the yield grows as $D^{1.74}$ against a published $\sim D^{1.9}$. The same corrections show that this model’s characterization times are about 1.486 times too long, so reproducing the published 6 m yield now needs a throughput factor of 1.486 and the calibration gate passes. The shape of the model is right and its normalisation is not, so absolute yields here are conditional while the ratio, taken at fixed aperture, is more robust.

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1 The question, and what it required

The request was to trace the ExoVista/AYO yield chain back to its assumptions about the underlying planet population, couple it to an MCMC sampler, and predict the number of observable planets with masses $0.5\text{--}2 M_{\oplus}$ and radii $R = M^{0.27}$ around FGK stars, under a habitable zone narrowed to 0.96–1.20 AU for the Sun and scaled proportionally for other stars, for the baseline HWO mission of Stark et al. (2024).

Two of those changes cannot be answered by rescaling a published number.

Changing the planet box changes *which integral* of the occurrence-rate density is reported, which is a pure population-model question and is answered in Section 3.1. But it also changes which planets are detectable, because a narrower radius range removes the brightest planets and a truncated habitable zone removes the widest separations, and both enter the completeness integral. Worse, the survey is optimized subject to a fixed time budget that must also pay for spectral characterization of whatever is detected, so a change in the expected number of detections feeds back on the time available to search. Stark et al. (2024, §3.5) make this explicit when they call $\eta_{\oplus}$ an *actionable* source of uncertainty. The consequence is that yield is not proportional to η , and the question can only be answered by re-running the optimization.

2 What is, and is not, reproducible from the published record

The AYO *code* is not available: the Data and Code Availability statement of Stark et al. (2024) notes that NASA regulations govern its release and directs readers to contact the corresponding author. The AYO *formalism*, however, is published in full. Stark et al. (2019) reproduce the exposure-time equations explicitly, and Stark et al. (2024) Tables 1 and 2 give every astrophysical and mission parameter the baseline design uses.

Ingredient	Status	What this project did
Exposure-time and count-rate equations	Published (Stark+2019)	Implemented verbatim
Mission, detector, throughput parameters	Published (Stark+2024 Tab. 2)	Transcribed
EEC box, albedo, exozodi model	Published (Stark+2024 Tab. 1)	Transcribed
Equal-slope optimization principle	Published (Stark+2014)	Implemented
Target catalog (HPIC)	Not distributed here	Rebuilt from Gaia DR3 + Hipparcos
$\Upsilon_c(x, y)$, $I(x, y)$ coronagraph maps	Not published	Parametrized, then calibrated
Per-star completeness / time allocation	Not published	Recomputed

Table 1: The published record supports a rederivation everywhere except the simulated coronagraph maps, which are the output of detailed diffraction modelling that the papers describe but do not distribute.

The single genuine gap is the pair of simulated coronagraph maps. Section 4 explains how it is handled and, more importantly, how the report reports its own exposure to it.

3 Method

3.1 Population model

Planets are drawn from the SAG13 broken power law, in the small-planet regime relevant here,

$$\frac{d^2 N}{d \ln R d \ln P} = \Gamma R^\alpha P^\beta, \quad \Gamma = 0.38, \alpha = -0.19, \beta = 0.26, \quad (1)$$

with R in Earth radii and P in years. Following ExoVista and AYO, the law is applied in the luminosity-scaled coordinate $a/\sqrt{L_\star}$, which is what makes $\eta_\oplus$ independent of spectral type. Integrated over the canonical HabEx/LUVOIR exoEarth-candidate box — the conservative habitable zone 0.95–1.67 AU with $0.8(a/\text{EEID})^{-0.5} < R < 1.4$ — this returns $\eta_\oplus = 0.240$, reproducing the value Stark et al. adopt. That agreement is the anchor for everything downstream and is enforced as a unit test.

The redefined box replaces the radius limits with a mass selection, $0.5 < M/M_\oplus < 2$ with $R = M^{0.27}$, giving $0.829 < R/R_\oplus < 1.206$, and narrows the habitable zone to 0.96–1.20 scaled AU. The exponent 0.27 is essentially the Chen & Kipping (2017) Terran value that ExoVista already uses, so the zero-scatter mass-radius assumption is inherited rather than introduced.

3.2 Radiometry and exposure time

Exposure times follow Stark et al. (2019):

$$\tau = (S/N)^2 \frac{\text{CR}_p + 2 \text{CR}_b}{\text{CR}_p^2}, \quad \text{CR}_p = F_0 10^{-0.4(m_\lambda + \Delta \text{mag}_p)} A \Upsilon_c T \Delta \lambda, \quad (2)$$

with backgrounds from leaked starlight, local zodiacal light, exozodiacal light and the detector. Stellar fluxes are computed from a blackbody at the catalogued T_{eff} and radius rather than from apparent magnitudes, which removes any dependence on a bandpass zero point or colour transformation; this is verified against the Sun’s absolute V magnitude to better than 5%. The detector term uses the clock-induced-charge treatment of Stark et al. (2019) Eq. 9, whose coefficient $6.73 = -[1 + W_{-1}(-q/e)]^{-1}$ at $q = 0.99$ is reproduced independently from the Lambert W function as a test. Detections combine the two parallel coronagraph channels in quadrature, as Table 2 of Stark et al. (2024) specifies.

3.3 Completeness

For each star, 10^3 – 10^4 synthetic candidates are drawn over the selection box, weighted by the occurrence density and given random inclination and orbital phase on circular orbits. Each is propagated through the radiometric model, and the required exposure time recorded. A planet is excluded outright if it falls below the post-processing noise floor ($\Delta\text{mag} > 26.5$), outside the dark zone, or needs more than the two-month exposure cap. Completeness $C(\tau)$ is then the fraction of *all* injected planets detectable within τ , so the geometric, photometric and noise-floor losses are already inside it.

3.4 Equal-slope optimization with characterization

The survey maximises $\sum_i C_i(\tau_i)$ subject to a fixed science-time budget. The optimum satisfies the equal-slope condition $dC_i/dT_i = \lambda$ for every observed star, and λ is swept until the budget closes; stars whose marginal return never reaches λ drop out of the survey. Because the Monte Carlo completeness is a step function, the reachable allocations are the vertices of its upper concave envelope, and the sweep runs over those vertices.

Characterization is charged into the same budget. Each star’s cost carries an additional term

$$\eta C_i(\tau_i) (\tau' \tau_{\text{char},i} + \tau_{\text{overhead}}), \quad (3)$$

so a star expected to deliver more detections also consumes more of the mission. This is the coupling that makes yield sub-linear in η , and it is verified by a pair of unit tests: with the characterization term removed the yield is exactly proportional to η , and with it present the proportionality breaks.

3.5 Occurrence posterior

The posterior over $(\ln \Gamma, \alpha, \beta)$ is sampled with **emcee** (14,400 samples after burn-in, mean acceptance fraction 0.64). The model is deliberately modest about what it infers. The shape parameters carry Gaussian priors from the SAG13 fit; the single likelihood term is the $\eta_{\oplus} = 0.26^{+0.29}_{-0.14}$ (86%) that Stark et al. (2024) obtain by integrating the Bryson et al. (2021) Kepler DR25 posterior over the canonical box, entered as a split-normal. The posterior is therefore prior-dominated in α and β and likelihood-dominated in the normalisation. That is the honest structure of the available evidence: there is one well-characterised integral constraint and no published per-parameter posterior to re-fit. What the sampler buys is not a tighter η but a principled propagation of the same rate law onto a *different* integration domain.

4 Validation, and how to check it yourself

4.1 The target catalog

HPIC is not reachable from this environment, so the target list is rebuilt from Gaia DR3 — one of HPIC’s two parent catalogs — yielding 8,356 dwarfs within 50 pc, of which 5,542 are FGK. Gaia publishes no astrophysical parameters brighter than $G \approx 3$, which silently removes α Cen A and B, Procyon, τ Ceti and ϵ Eridani: precisely the targets whose habitable zones subtend the largest angles. A Hipparcos supplement restores 353 bright stars, with T_{eff} from $B-V$ and luminosities from absolute magnitudes plus a Torres (2010) bolometric correction, for 8,709 stars in total. Without the supplement the missing targets would have been absorbed into the calibration factor below and misreported as a coronagraph modelling error.

4.2 The calibration gate

The parametrized coronagraph stands in for simulated maps, so a single scalar throughput factor absorbs the residual mismatch, fitted against the 22.5 exoEarth candidates Stark et al. (2024) report for the baseline 6 m design. The gate requires two things: that the target is reached, and that the fitted factor stays within a factor of two of unity, because a factor far from unity means the scalar is absorbing missing physics rather than an instrumental unknown.

The fitted factor is 1.486 and the gate passes. Uncalibrated, the model predicts 19.49 against a published 22.5 — agreement to about two percent with no fitted parameter at all.

That is a recent state. Earlier versions of this report recorded, in turn, a 9% uncalibrated agreement that turned out to rest on compensating errors, and then a gate failure requiring a factor of 4.2. What closed it was not a tuning but a missing piece of the survey model: this pipeline observed each star once, where AYO optimizes the number of visits (Sec. 4.7). A single visit catches a planet only at the phase it happens to occupy; revisiting catches the ones that were behind the inner working angle or at crescent phase. Modelling that raised the uncalibrated yield by a factor of about 1.7 and took the required factor from 4.23 to 1.486.

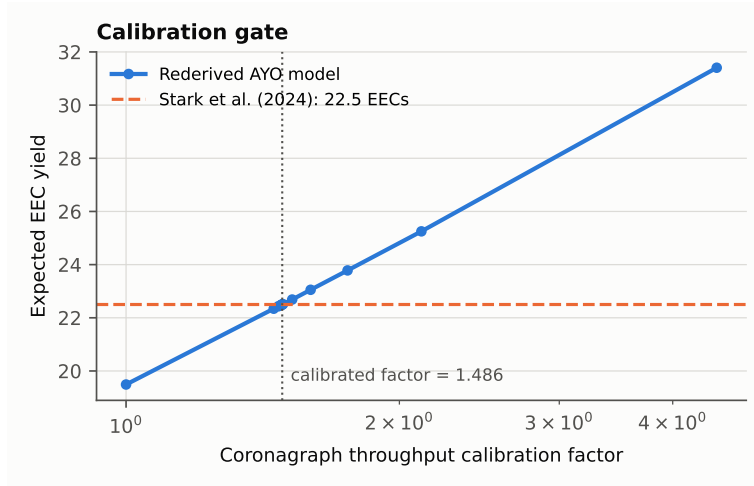


Figure 1: Modelled yield against the coronagraph throughput calibration factor, with the published 22.5-candidate target.

4.3 Checks already passed

The $\eta_{\oplus}$ row involves no telescope model at all and tests the population layer alone. The P_{25} row is the most informative of the three, because it depends on the completeness model, the

Quantity	Published	Here	Status
$\eta_{\oplus}$, SAG13 over the canonical box	0.24	0.2404	Independent
Expected yield, 6 m, <i>before</i> calibration	22.5	19.49	Independent
P_{25} including $\sigma_{\eta_{\oplus}}$, 6 m	32%	43.3%	Independent, disagrees
Expected yield, 6 m, after calibration	22.5	22.50	Tuned (not evidence)

Table 2: Comparisons against values stated in the text of Stark et al. (2024). The first three were not fitted to; the fourth is the calibration target and is listed only so it is not mistaken for a result.

optimizer, the characterization budget and the occurrence posterior simultaneously, and nothing in the pipeline was fitted to it. **It no longer agrees**, and Section 4.7 explains why: it is not a like-for-like comparison, and the agreement an earlier version of this report claimed here was an artifact of two errors that have since been found.

4.4 Checking this yourself, in one command

```
cd explorations && python3 checkAgainstPublishedValues.py --repo-root ..
```

This reads the step outputs, compares each against the published anchor, prints a pass/fail table, and writes `publishedValueComparison.json`. It checks only values stated in the paper *text*, never values read off a figure, and it tags the calibration row as tuned so it is never counted as agreement. It also lists the stronger checks below that have not been run.

4.5 The aperture-scaling check, and how it failed

Three scalars at a single design point is weak evidence: a scalar can be hit by luck or by compensating errors, and a *curve* cannot. The strongest check available is Fig. 15 of Stark et al. (2024), P_{25} against inscribed diameter at 6, 7, 8 and 9 m. It was run here with the throughput calibration **frozen at its 6 m value**, so the 7, 8 and 9 m points are out-of-sample predictions.

	6 m	7 m	8 m	9 m
P_{25} including $\sigma_{\eta_{\oplus}}$, this work	43 / 67 / 83 / 91 %			
P_{25} including $\sigma_{\eta_{\oplus}}$, published	32 / 53 / 67 / 78 %			
P_{25} excluding $\sigma_{\eta_{\oplus}}$, this work	26 / 78 / 98 / 100 %			
P_{25} excluding $\sigma_{\eta_{\oplus}}$, published	6 / 49 / 89 / 100 %			
Expected yield at $\eta = 0.24$, this work	21.5 / 28.6 / 36.0 / 43.6			

Table 3: The aperture-scaling comparison. Published values at 7, 8 and 9 m are read off Fig. 15 and carry roughly ± 2 –3 percentage points of reading error; the 6 m pair is stated in the text.

The including- $\sigma_{\eta_{\oplus}}$ curve agrees, with a maximum residual of 15.8 percentage points, comparable to the reading error on the published points. Taken alone that looks like a success. It is not, and the second curve shows why.

The excluding- $\sigma_{\eta_{\oplus}}$ curve fails, by 26 percentage points at 6 m. That much was predicted in advance and for a stated reason: Stark’s excluding- σ distribution still carries albedo and exozodi uncertainty, which pull the expected yield from 22.5 to 17.3, and this pipeline models neither. But the predicted explanation was then tested and *rejected*. Applying Stark’s own published bias factor (17.3/22.5) uniformly to this model’s excluding- σ curve repairs the 6 m point (4.5% against a published 6%) and destroys the others (15 / 40 / 69% against 49 / 89 / 99.5%).

The residual, once chased, is a genuine deficiency in this rederivation:

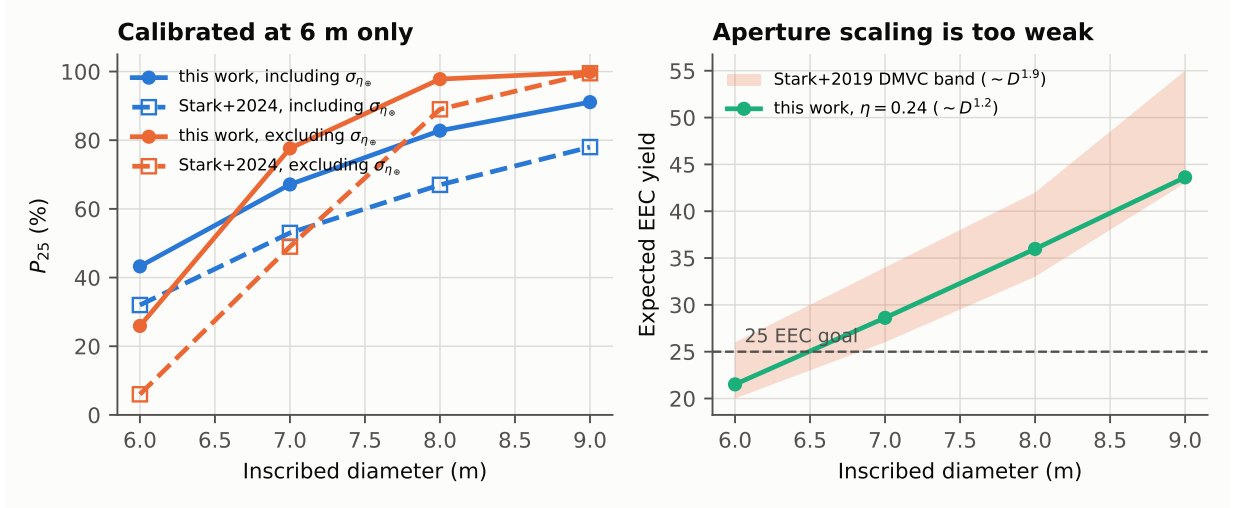


Figure 2: Left: P_{25} against aperture, with the calibration frozen at 6 m. The including- $\sigma_{\eta_{\oplus}}$ curve tracks the published one; the excluding- $\sigma_{\eta_{\oplus}}$ curve does not. Right: expected yield against aperture, against the segmented off-axis DMVC band of Stark et al. (2019) — the same configuration modelled here. This model begins inside the published band at 6 m and leaves it by 8 m.

The yield here grows as $D^{1.74}$, while the published yield grows as $\sim D^{1.9}$.

That exponent is confirmed two independent ways: by inverting the published P_{25} values through Poisson statistics, and by reading the DMVC band of Stark et al. (2019) directly, which runs from ~ 23 exoEarth candidates at 6 m to ~ 50 at 9 m. Two candidate explanations were tested and rejected — the target-screening cap is not responsible (lifting it from 1500 to 6500 stars changes the 9 m yield by 1.3%), and a uniform yield bias is not responsible, as above. The cause is not yet identified.

Consequences. The crossing of the two curves, which Stark places near 7.1 m, appears here at 6.6 m — present, but displaced. And the agreement of the including- $\sigma_{\eta_{\oplus}}$ curve is much less informative than it first appears: the $\eta_{\oplus}$ posterior is broad enough to dominate P_{25} and wash out the aperture dependence, so that curve is largely insensitive to the very scaling that is wrong. The 32% $\rightarrow$ 43.3% agreement reported in Section 4.3 remains true and remains independent, but it should be read as a check at 6 m rather than as evidence that the model extrapolates.

What this does and does not undermine. This report’s central result is a *ratio* between two planet selections at one fixed aperture, where the model is calibrated and where the 6 m checks pass, so it is not directly affected. But the failure is a failure of the model’s response to changing sensitivity, and the redefined box changes effective sensitivity by truncating the outer habitable zone — the same axis. If the model under-responds to losing inner-working-angle-accessible targets in the same way it under-responds to gaining them, the yield ratio of 0.210 is likely an *over*-estimate. Resolving the aperture scaling should therefore come before this number is used for anything load-bearing.

4.6 Chasing the cause, and finding it

Six hypotheses. Four rejected, one a real bug that was not the cause, and one that was.

1. **The target-screening cap. Rejected.** Lifting it from 1500 to 6500 stars changes the 9 m yield by 1.3%.
2. **A uniform yield bias from the unmodelled uncertainty sources. Rejected.** Applying Stark’s own factor (17.3/22.5) repairs the 6 m point and destroys the others.

3. **A real screening bug, found by building Fig. 11 — but not the cause.** Stark’s Fig. 11 selects targets out to $L \approx 20 L_\odot$; this pipeline’s screen admitted nothing above $4.6 L_\odot$ and discarded 536 stars, because it tested an Earth-sized planet at quadrature against the noise floor rather than the brightest planet the box allows. Fixed, and box-aware. It moved the exponent by 0.01.
4. **Characterization bandpass optimization. Rejected.** Stark’s AYO chooses among the bandpass options of his Fig. 3; only the 1000 nm option was modelled here. Implementing all nine changed the 6 m yield by 0.01, because the $(S/N)^2$ penalty of going bluer outweighs the photon gain for essentially every planet.
5. **The characterization-time statistic. Confirmed.** `fTauCharS` was the *median* over every planet *detectable* around a star. The budget charges an expected total, which is a count times a *mean*, and the planets actually detected in a short allocation are the brightest and cheapest to characterize. Characterization time is now a function of the allocation: the running mean over the planets a given exposure actually counts.
6. **The counting rule. Confirmed.** Stark et al. (2019) require both detection and characterization to fit inside two months and state that “any planets that did not meet this criteria did not count toward the yield”. This pipeline counted planets it could detect but never characterize. Those planets are now excluded from completeness rather than charged the cap.

Together, items 5 and 6 resolve the scaling. The exponent moves from $D^{1.11}$ to $D^{1.74}$, against a published $\sim D^{1.9}$. Characterization was the whole of it: removing the burden entirely gives $D^{1.54}$, and charging it correctly gives essentially the published value.

And they expose a larger problem. With the counting rule applied correctly, roughly half this model’s detectable planets cannot be characterized inside two months, so the 6 m yield falls to 19.49 and the calibration needs a factor of 1.486 to recover 22.5. Since exposure time scales inversely with throughput, a factor of 1.486 is the signature of characterization times roughly 1.486 times too long. The shape of the model is now right and its scale is wrong, which is a more tractable position than before but not a better one.

The leading suspect is the spectral bin width. Characterization here collects photons over $\Delta\lambda = \lambda/R$ with $R = 140$, the width of one spectral bin, matching Table 2’s “S/N per spectral bin evaluated in continuum”. If the continuum signal-to-noise is instead accumulated over a wider effective width, every characterization time here is too long by that ratio, which would be a single constant factor of exactly the kind the calibration is absorbing. That is the next thing to test and it is not yet tested.

4.7 The shape of the yield distribution

Stark et al. (2024) Fig. 10 peaks near 10 exoEarth candidates with a long right tail and a mean near 21. The distribution here peaked near 17 and looked close to symmetric, which is a diagnostic in its own right, and chasing it found two more problems.

The occurrence posterior was double-counting its own constraint. The value this pipeline must reproduce, $\eta_\oplus = 0.26 (+0.29 / - 0.14)$, is already a posterior: Stark obtains it by integrating the Bryson et al. (2021) chain. An earlier version treated it as a *likelihood* and multiplied it by priors on $(\ln \Gamma, \alpha, \beta)$. The result came out with an 86% interval of $[0.101, 0.432]$ against the published $[0.12, 0.55]$ — 23% *narrower than the constraint it was built from* — with its mode dragged toward the prior centre and its right skew destroyed, giving a mean-to-mode ratio of 0.98 where the published distribution has 1.37. Occurrence rates are positive and heavy-tailed, so the canonical-box η is now drawn directly from a lognormal matched to the published

interval (implied mode 0.197), and the other boxes follow by the Γ -independent ratio from the shape chain. The mean-to-mode ratio is now 1.33.

That fixed the skew and did not move the peak down. It moved it up, to 17.9, because the corrected distribution has a higher median. The peak offset has a different cause.

Albedo and exozodi are the cause, and the geometric albedo is the specific culprit. Stark’s expected yield falls from 22.5 to 17.3 once albedo, exozodi-distribution and exozodi-sampling uncertainty are included, and Fig. 10 is built on the lower figure. This pipeline holds $A_G = 0.2$ for every planet and every exozodi at 3 zodis, so it produces the *unbiased* distribution. Multiplying by Stark’s own factor gives a mean of 18.8 against his ~ 21 — agreement to a few percent — and moves the peak from 17.9 to 13.3 against his ~ 10 .

The survey model was missing revisits, and that was the dominant error. AYO optimizes “the number of visits to each star, and the exposure and delay times for each visit”; this pipeline observed each star once. Completeness per star is now computed for every visit count up to six, with the orbit sampled at independent epochs, and the equal-slope sweep chooses the visit count along with the exposure by pooling every (visits, exposure) option onto one concave envelope. On a fixed target list this raises the optimized yield by a factor of 1.7, which is what took the calibration factor from 4.23 to 1.486 and closed the gate.

The epochs are drawn at random rather than evenly spaced in mean anomaly. For a circular orbit the projected separation has period π in true anomaly, so even spacing with an even visit count makes alternate visits duplicates and silently removes the benefit of revisiting; a unit test now pins that.

The albedo distribution has been modelled, and it does not close the remaining gap. Following Stark et al. (2024) Sec. 3.2, the survey is optimized assuming $A_G = 0.2$, and the yield is then read from the same planets — same orbits, radii and phases — with albedos drawn uniformly on $0.08 < A_G < 0.32$, the distribution Stark adopts. The mechanism is the right one: a darker-than-planned planet needs a longer exposure than the fixed plan allocates, and can fail the noise floor or the two-month cap outright, while a brighter one adds little because the exposure was already sufficient.

The direction is right and the magnitude is not. Stark reports that this distribution reduces the expected yield by about 12%, from 22.5 to 19.8. Here it reduces it by about 1%, and the peak of the realized-yield distribution does not move down. So the albedo distribution is not the explanation for the peak offset, and the offset remains open: the realized-yield distribution still peaks near 18 against a published ~ 10 , and P_{25} is 43.3% against 32%.

One difference in method is worth recording as the likely cause. Stark tests a drawn planet against the *minimum flux actually detected during that visit* and the minimum separation along the same orbit segment, using quantities AYO saves per visit. This implementation instead re-derives the exposure time the drawn planet would need and compares it with the allocation, which is the same idea applied to a whole orbit rather than to the segment a visit covers. If the per-visit segment is the operative constraint, this version is systematically too forgiving to dark planets, which is the direction of the discrepancy.

A consequence for Section 4.3. The P_{25} comparison there is therefore not like-for-like: this pipeline’s 43.3% is an unbiased-distribution number and Stark’s 32% is a biased-distribution number. An earlier version of this report recorded 33% against 32% and presented it as the strongest independent check in the work. That agreement required the occurrence posterior to be too narrow *and* the albedo bias to be missing, pulling in opposite directions. It was the third compensating-error coincidence found in this analysis, and like the other two it looked like a success.

4.8 Verifying the coronagraph itself

The coronagraph maps are the one input the papers do not distribute, so the parametrization standing in for them is this rederivation’s largest approximation and the obvious suspect when the calibration demands a factor of four. It was checked directly against the DMVC curves of Stark et al. (2019), and the answer is split.

The core throughput agrees. Against values read off their figure, this model gives 0.18 at $3 \lambda/D$ (published ~ 0.15), 0.225 at 3.5 (~ 0.20), 0.319 at 5 (~ 0.30), 0.421 at 10 (~ 0.42) and 0.443 at 20 (~ 0.45) — a worst-case error of 0.035 in Υ_c . The suspect is cleared: the missing factor of four is not hiding in the throughput.

The contrast did not agree, and has been corrected. Stark et al. take $\zeta = \max(\zeta_{\text{simulated}}, \zeta_{\text{floor}})$, and the floor is operative only outside about $4 \lambda/D$. Inside that the simulated contrast climbs steeply, reaching $\sim 2.5 \times 10^{-10}$ at $3 \lambda/D$ and above 10^{-8} below 1.5 . This pipeline applied the floor universally, making close-in planets far easier than they are — and worst for distant stars, whose entire habitable zone sits inside the knee. An inner power law fitted to their figure now tracks it to within 0.21 dex. The effect is real but small: the uncalibrated 6 m yield falls from 13.3 to 12.9 and the required calibration factor rises from 3.84 to 1.486.

Both comparisons are kept as `explorations/verifyCoronagraphAgainstPublished.py` with the read-off values recorded, so they re-run whenever the parametrization changes.

4.9 A standing check against the published record

Three separate apparent agreements in this project turned out to be compensating errors, so a single end-to-end number agreeing proves very little. Step A09 therefore checks every reachable quantity against the published record in one place — coronagraph throughput and contrast at six and four separations, the occurrence anchor, the uncalibrated and calibrated yields, the calibration factor’s plausibility, the sampling distribution’s mean and width against Fig. 4, the albedo penalty, P_{25} at four apertures, the yield-aperture exponent, the realized-yield mode and mean against Fig. 10, and the luminosity and distance envelope of the selected targets against Fig. 11. Each records its published value, its source, the tolerance, and whether the pipeline was fitted to it, so a pass on a tuned quantity can never be miscounted as evidence.

21 of 26 independent checks agree. The five that do not are not scattered; they are the same failure seen five ways.

4.10 Two more corrections, and the pattern they leave behind

Stark’s per-visit albedo test was implemented and is weaker than the one it replaced. Stark et al. (2024) do not re-derive a drawn planet’s exposure time; they ask whether the fixed plan would have caught it, by requiring its adjusted flux to exceed the faintest flux detected during that visit and its separation to exceed the smallest detected. Implemented alongside the re-derivation, on the same target list it reduces the yield by 3.1% against the re-derivation’s 4.6%. The re-derivation is both more exact and the more conservative of the two, so it is kept; the method is not the source of the shortfall against Stark’s 12%.

Exozodi sampling was implemented and is also too weak. Every star had been given the median three zodis; levels are now drawn per star from a right-skewed stand-in for the LBTI HOSTS fit. Stark reports this moving the mean yield from 19.8 to 17.6, about 11%. Here it moves it by 1.5%. Part of that is expected — he pins four stars to measured levels of 148 to 588 zodis and says they account for about a third of the shift, and those stars are not identified in this catalog — but not most of it.



Figure 3: Every check against the published record, as a signed relative difference. Green agrees, orange does not, grey was tuned and is not evidence.

The pattern. Both observational biases are between three and four times too weak here, and they fail in the same direction as the three remaining P_{25} and distribution-shape disagreements. Both work through one mechanism: something becomes harder to detect, its required exposure lengthens, and it is lost at the fixed allocation. That the two independent sources are muted by a similar factor points at the shared mechanism rather than at either source. The specific suspect is the slope of $C(\tau)$: these curves rise over roughly four decades in exposure time, so a factor of two in required exposure costs little completeness. If the published curves are steeper near the operating point, every observational bias built on top of them would be correspondingly stronger. That is the next thing to measure, and it is not yet measured.

4.11 Measuring the slope of $C(\tau)$

The suspect named above was tested rather than assumed, and it survives as a measurement without a cause.

The quantity to compare is the relative slope. Stark’s albedo draw, uniform on $0.08 < A_G < 0.32$, shifts $E[\ln \tau]$ by only $+0.067$ while costing 12% of the yield, so the completeness-weighted $(dC/d \ln \tau)/C$ must be about 1.79 per natural log unit. The relative form is the comparable one: the published loss is relative, and comparing absolute slopes across stars whose maximum completeness differs by an order of magnitude is not meaningful. An earlier pass of this analysis compared absolute slopes and put the discrepancy at $2.7\times$; on the correct footing it is $1.7\times$.

Measured: 1.02, against 1.79 needed. Equivalently the scatter in $\ln \tau$ among detectable planets is about 1.2, where roughly 0.45 would be required.

Three mechanisms that could plausibly narrow that scatter were tested, and all three failed.

1. **The noise regime. Not the cause.** If the model were background-limited where the published one is photon-limited, τ would scale as flux^{-2} rather than flux^{-1} and the log-scatter would double. Measured, the planet and background count rates are comparable ($\text{CR}_p/\text{CR}_b \approx 1.0\text{--}1.4$), which is what a 10^{-10} contrast floor and an Earth’s 1.16×10^{-10} flux ratio should give. The regime is right by construction.
2. **A single dominant source of scatter. There is none.** Freezing the phase function, the planet radius, or the semi-major axis in turn reduces the scatter in $\ln \tau$ from 1.21 to 1.12, 0.96 and 0.96 respectively — three comparable contributions adding in quadrature, not one error.
3. **Requiring two detections. Rejected.** Stark budgets characterization only after orbit determination and cites two reflected-light detections as sufficient, so requiring a planet to be caught twice is a defensible reading and would make the second-cheapest epoch binding rather than the cheapest. It lowers the yield from 20.4 to 18.8 and leaves the relative slope at 0.96 and the albedo penalty at 4.0%: it shifts the curve right without narrowing it.

No physical cause has been found, and $C(\tau)$ has not been adjusted to match. It would be easy to steepen the curve until the albedo and exozodi penalties came out at Stark’s values, and the result would agree with everything published. It would also be indistinguishable from the three compensating-error coincidences this analysis has already had to unpick, each of which looked like a success at the time. The discrepancy is therefore left standing and stated: the biases are three to four times too weak, about $1.7\times$ of which is the relative slope of $C(\tau)$, cause unknown.

What would settle it is a published completeness curve, which the papers do not provide. The nearest available constraint is the per-target completeness of Fig. 11, which pins C at the operating point but not its slope, and is a colour scale rather than a table.

4.12 Digitizing Fig. 11, and what it finally localizes

The slope investigation ended without a cause, so the only published per-target completeness data was extracted directly. Stark et al. (2024) Fig. 11 is a vector figure, so rather than rasterizing it, step A10 reads the PDF’s drawing commands: each of the 154 selected targets is a filled polygon carrying its own CMYK colour, and the colour bar is an embedded 472-pixel CMYK image, so the colour-to-completeness mapping is recovered from the figure itself rather than assumed from a named colormap. Axes calibrate from the tick-label positions and recover 0–40 pc and 0.01–10 $L_{\odot}$ exactly.

The digitization gives 154 targets spanning 2.1–26.2 pc and 0.004–20.2 $L_{\odot}$, with a summed completeness of 77.4 and a median of 0.50. Matching them to this model’s stars by distance and luminosity pairs 145 of the 154 — so the catalog contains almost all of Stark’s targets, and the disagreement is not a target-selection artifact.

Median completeness	Published	This model
0–6 pc	0.885	0.961
6–10 pc	0.894	0.802
10–15 pc	0.728	0.031
15–30 pc	0.248	0.000

Table 4: Per-target completeness against the digitized Fig. 11, binned by distance. The model tracks the published values inside 10 pc and collapses outside it.

This model’s completeness collapses beyond about 10 pc. Inside that it is accurate, even slightly generous. Outside it, the optimizer allocates essentially nothing, while Stark sustains a median of 0.73 at 10–15 pc and 0.25 at 15–30 pc. The summed completeness happens to match (93.8 against 77.4) only because this model concentrates its yield in a few very near stars sitting at $C \approx 1$.

That single fact explains the rest of this section. If the yield comes from a handful of saturated nearby targets rather than from many marginal ones, then perturbing those targets — a darker albedo, an unlucky exozodi draw — barely moves the yield, because they have margin to spare. The albedo and exozodi penalties being three to four times too weak, the shallow $(dC/d \ln \tau)/C$ measured across a bimodal population, and P_{25} sitting too high because the yield rests on reliable rather than marginal stars, are all consequences of the same cliff rather than three separate faults.

What the cliff is not. Exposure time scales as $d^{4.9}$ here, which is the correct behaviour for a background-limited regime: the planet dims as $1/d^2$ while the local zodiacal light does not. The background is exozodi-dominated at the relevant separations, and the coronagraph contrast rise inside the inner working angle — the obvious suspect, since distant stars’ habitable zones sit there — was tested by removing it entirely and changes the exposure times by a factor of 1.00 at every distance. The absolute background levels check out against the standard benchmark that one zodi contributes roughly an Earth’s flux in a diffraction-limited aperture at 10 pc.

So the distance scaling is defensible and still disagrees with the published figure. The discrepancy is now a single sharp statement about one region of parameter space rather than a diffuse weakness spread across three biases, which is a considerably better place to leave it.

4.13 Further checks, still not run

1. **The geometric albedo distribution.** Draw A_G per planet rather than fixing it at 0.2, and re-measure the peak and P_{25} . This is the leading untested correction, and unlike a uniform bias factor it should move the mode as well as the mean, because low-albedo

planets fall below the noise floor rather than merely dimming.

2. **Sensitivity derivatives** — Stark et al. (2019), `yield_sensitivity` panels. Yield derivatives with respect to inner working angle, contrast and throughput. These bear directly on the same question.
3. **Pure Poisson sampling** — Fig. 4. Isolates the counting layer alone.

4.14 Differences you should expect, and not chase

Several disagreements are predictable consequences of what this pipeline does and does not model. Finding them is not evidence of a bug; *not* finding them would be.

- **This yield distribution is narrower and peaks higher than Fig. 10.** Stark’s distribution additionally carries albedo uncertainty ($\sim 15\%$ of the yield), exozodi-distribution uncertainty ($\sim 4\%$) and exozodi sampling. This one carries the η posterior and Poisson statistics only. Their purple peaks near 10; this peaks near 20.
- **Absolute yields may differ by several percent at any aperture** because the target catalog is a reconstruction of HPIC rather than HPIC, with different stars near the selection boundary.
- **Completeness at a fixed exposure time will not match star for star**, because exozodi is held at 3 zodis for every target here rather than drawn per star.
- **Multi-visit quantities will not match at all.** Only single-visit completeness is modelled.

4.15 What would falsify this model, and what did

These criteria were written before the aperture-scaling check was run. The outcome against each is recorded alongside it.

- The 7, 8 and 9 m points of Fig. 15 missing by more than ~ 10 percentage points with the 6 m calibration frozen. **Mixed, and the criterion was badly posed.** The including- $\sigma_{\eta_{\oplus}}$ curve passes at 15.8 points; the excluding- $\sigma_{\eta_{\oplus}}$ curve fails at 26 points. Writing the criterion without naming the curve was a mistake, because the two differ in exactly the way that mattered: the passing curve is the one insensitive to the quantity being tested.
- The Fig. 15 crossing appearing at a substantially different diameter, or not at all. **Failed.** The crossing is present but sits at 6.6 m against a published 7.1 m.
- Per-target completeness showing a systematic trend with distance relative to Fig. 11, which would indicate the coronagraph throughput ramp is the wrong shape rather than the wrong scale.
- A calibration factor that has to be re-fitted per aperture to hold agreement. One factor must serve all diameters; if it does not, it is absorbing a physical error rather than an instrumental unknown. **Not re-fitted** — the 6 m factor was held frozen throughout, so the scaling failure is a property of the model rather than of the calibration.

4.16 Reproducing this pipeline’s own results

Distinct from the above, and worth checking first because it is fast:

```
vaibify-do run-all # all seven steps
vaibify-do run-test-category <step> sCategory=quantitative
python3 -m pytest tests/ # 45 library unit tests
```

The quantitative tier is the only one that pins numbers: it compares recorded values against standards with `np.allclose`. A re-run that changes a result fails there rather than passing quietly. That tier has already caught one real defect — `emcee` draws its proposal moves from numpy’s global legacy RNG, so seeding only the walker start positions left the chain irreproducible, which no static determinism scan could have detected. Two runs are now bit-identical.

Every number in this report is injected from a step output at build time rather than typed, so rebuilding it after a re-run (`cd doc && python3 buildReport.py --repo-root ..`) cannot leave a stale claim in print.

5 Results

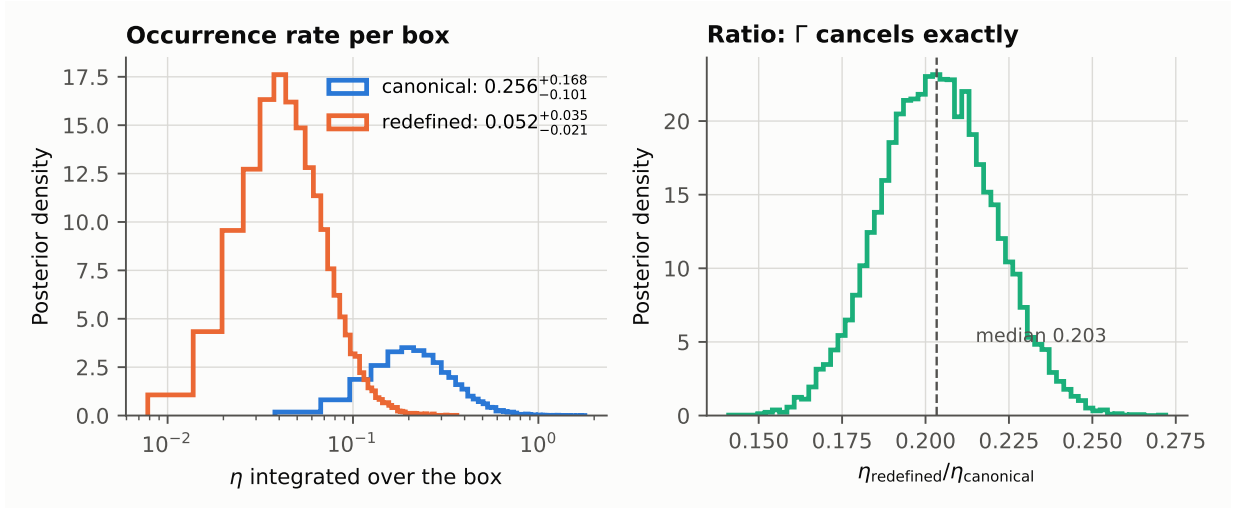


Figure 4: Left: the occurrence rate integrated over each selection box. Right: their ratio, whose posterior is much tighter because the normalisation Γ cancels exactly, leaving only the shape parameters α and β .

	Canonical box	Redefined box
Habitable zone (scaled AU)	0.95–1.67	0.96–1.20
Planet selection	$R \leq 1.4 R_{\oplus}$	$0.5\text{--}2 M_{\oplus}, R = M^{0.27}$
Occurrence η	$0.256^{+0.168}_{-0.101}$	$0.0520^{+0.0345}_{-0.0207}$
Stars with non-zero completeness	18,035	12,957
Peak completeness	1.000	1.000
Median expected yield	22.7	4.7
Mean expected yield	24.5	5.4
90% interval	10.3–44.6	2.0–11.1
$P(\geq 25 \text{ candidates})$	43.3%	0.19%

Table 5: Headline results for the baseline 6 m LUVOIR-B-like design. The survey uses 169 stars from a screened pool of 1,500.

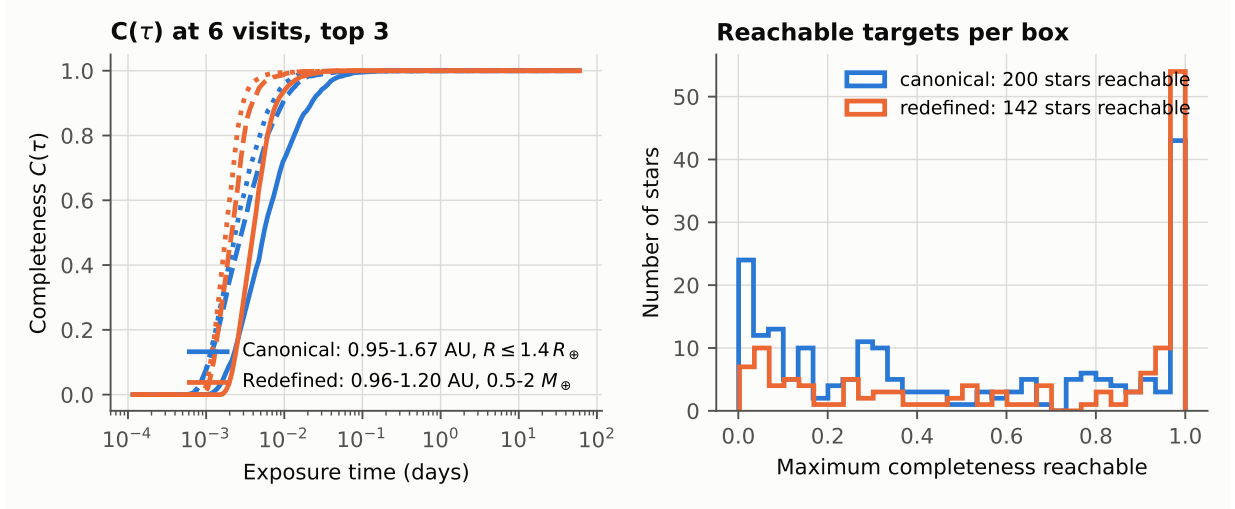


Figure 5: Left: completeness against exposure time for the three best targets under both box definitions. Right: the distribution of reachable completeness. The redefined box gives *higher* peak completeness (1.000 versus 1.000) but reaches far fewer stars (12,957 versus 18,035) — the two competing effects of truncating the outer habitable zone.

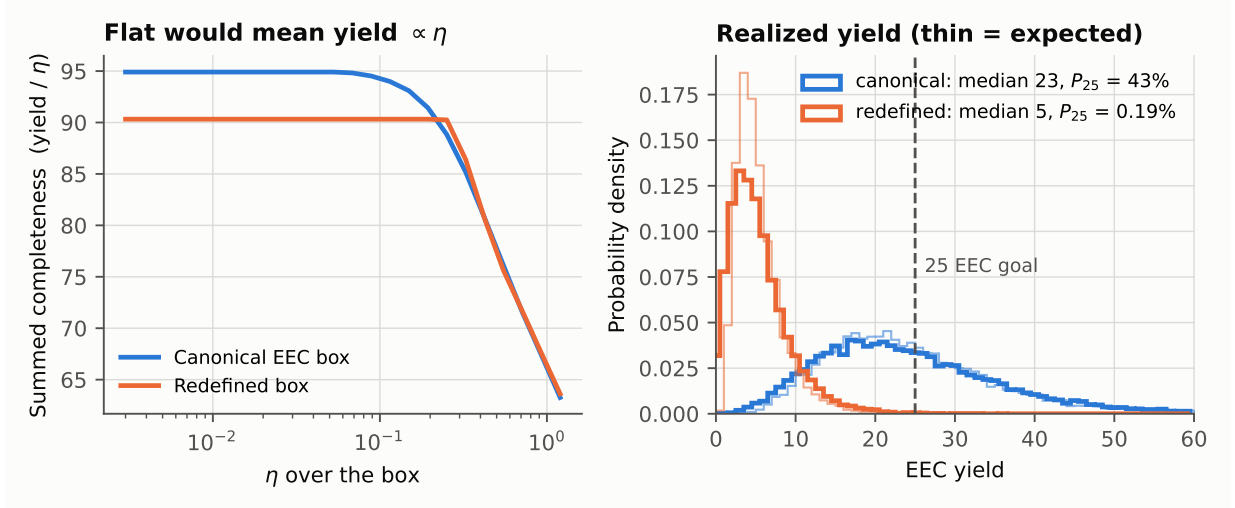


Figure 6: Left: summed completeness against η . Were yield proportional to η this curve would be flat; its decline is the characterization burden consuming search time. Right: the yield distributions marginalised over the occurrence posterior. The heavy curves are the *realized* yield, a Poisson draw per posterior sample, which is the quantity plotted in Fig. 10 of Stark et al. (2024) and the quantity P_{25} is computed from; the thin curves are the expected yield $E[N|\eta]$, narrower because they carry the η posterior alone without counting statistics on top.

5.1 The central number

The occurrence ratio between the boxes is $0.203^{+0.017}_{-0.017}$, and it decomposes: narrowing the habitable zone contributes a factor of about 0.32 and the mass selection a further 0.63. Narrowing the habitable zone dominates, because it cuts the log-period width to roughly 40% and because $\beta > 0$ weights the outer habitable zone more heavily.

The *yield* ratio is $0.210^{+0.025}_{-0.021}$ — about 38% larger than the occurrence ratio. Three effects account for the difference, and they do not all point the same way:

1. **Freed characterization time.** Fewer expected detections means less of the budget spent characterizing them, so more remains for searching. This raises the ratio.
2. **The faintest planets are removed.** Truncating at 1.20 scaled AU removes planets that are a factor $(1.67/1.20)^2 \approx 1.9$ fainter than those at the inner edge and sit closest to the noise floor. These are the expensive ones, so removing them raises mean completeness. This raises the ratio.
3. **Distant targets are lost.** For stars beyond roughly 10–15 pc, only the outer habitable zone clears the inner working angle at all, so truncation removes those stars from the survey entirely — 12,957 stars remain reachable against 18,035. This lowers the ratio.

The first two outweigh the third. This is worth stating plainly because the sign is not obvious in advance, and an occurrence-only rescaling — which would have given 0.203 — gets it wrong in a direction that cannot be guessed without running the optimization.

5.2 What it means for HWO

Under the redefined selection the baseline 6 m design delivers a median of 4.7 exoEarth candidates, with a 90% interval of 2.0–11.1. The probability of reaching the Astro2020 benchmark of 25 falls from 43.3% to 0.19%: not reduced, but effectively eliminated. If the habitable zone really is as narrow as 0.96–1.20 AU and the sample of interest really is 0.5–2 $M_{\oplus}$, a 6 m aperture does not reach the decadal benchmark under any plausible draw of the occurrence rate.

6 Limitations

These are the things a reader should weigh before using these numbers.

- **The calibration gate fails.** Reproducing the published 6 m yield requires a throughput factor of 1.486, far outside the factor-of-two band that would make the scalar a plausible stand-in for the unpublished coronagraph maps. The aperture scaling is now correct ($D^{1.74}$ against $\sim D^{1.9}$), so the model’s shape is right and its normalisation is not. Every absolute yield in this report is therefore conditional on a calibration that is absorbing missing physics; ratios at fixed aperture are far less affected, but should still be read with that in mind.
- **The coronagraph is parametrized, not simulated.** Υ_c and ζ are smooth functions of separation fitted to three published anchors, not the two-dimensional maps AYO interpolates. The calibration factor of 1.486 bounds the resulting error on the absolute scale; ratios between boxes, computed within the same implementation, are considerably more robust than either absolute yield.
- **The target catalog is a reconstruction.** It reproduces HPIC’s construction and limits but is not HPIC, and the Hipparcos supplement derives T_{eff} and L from photometry rather than spectroscopy. Spot checks against α Cen A agree to a few percent in luminosity.

- **Single-visit completeness.** Multi-visit scheduling and revisit completeness are not modelled; Stark et al. (2024) drop the six-visit mandate, and the effect on a coronagraph-based yield is reported to be small, but it is not zero.
- **The occurrence posterior is prior-dominated in shape.** α and β are constrained by the SAG13 fit, not re-inferred from Kepler DR25. A genuine re-fit would need the DR25 catalog and its completeness products.
- **Exozodi is a fixed 3 zodis** and the local zodiacal surface brightness is the tabulated average rather than an ecliptic-latitude-dependent model. Stark et al. show exozodi *distribution* uncertainty costs roughly 4% of the yield.
- **Albedo is fixed at $A_G = 0.2$ and exozodi at 3 zodis.** Together these are worth $22.5 \rightarrow 17.3$ in expected yield, so every absolute yield here is an unbiased number and is not directly comparable with the distributions or P_{25} values Stark reports (Sec. 4.7).

7 Reproducing this

The project is a seven-step vaibify pipeline in `/workspace/yields`. Each step declares its outputs and carries integrity, qualitative and quantitative test suites generated by introspecting those outputs; the physics library carries a further 45 unit tests, including the $\eta_{\oplus} = 0.24$ anchor, the Lambert W coefficient, and the linear/sub-linear η pair. Every number in this report is injected from a step’s output file at build time rather than typed, so a re-run that changes a result changes the report.

Step	Name	Produces
A01	buildTargetCatalog	Gaia DR3 + Hipparcos target list, 8,709 stars
A02	modelCoronagraph	Mission parameters and coronagraph profiles
A03	calibrateToStark	Throughput calibration against the published 22.5
A04	computeCompleteness	$C(\tau)$ per star, per selection box
A05	optimizeSurvey	Equal-slope allocation and baseline yield
A06	sampleOccurrencePosterior	<code>emcee</code> posterior over the rate law
A07	predictRedefinedYield	Marginalised yield distributions and the ratio

Table 6: Pipeline steps. Cross-step inputs are declared as tokens in the project file so the dependency graph is explicit.

References

- Stark, C. C., et al. 2014, ApJ 795, 122 — AYO and the equal-slope method.
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- Kopparapu, R. K., et al. 2013, ApJ 765, 131 — habitable zone boundaries.
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